# Exercise 5

1. Evaluate the performance of two classification models, $M_1$ and $M_2$. Table 1 shows the posterior probabilities for the positive class obtained by applying the models to the test set. Plot the ROC curve for both $M_1$ and $M_2$ and calculate the area under the ROC curve (AUC). Explain which model is better.

2. For model $M_1$, suppose the cutoff threshold is set to be $t = 0.5$, i.e., any test instances whose posterior probability is greater than $t$ will be classified as positive. Compute the precision, recall, and F1-measure for the model at this threshold value.

3. Repeat the analysis for task 2 using the same cutoff threshold on model $M_2$. Compare the F1-measure results for both models. Which model is better? Are the results consistent with the conclusions drawn from the ROC curves in task 1?

4. Repeat task 2 for model $M_1$ using the threshold $t = 0.1$. Which threshold is better, $t = 0.5$ or $t = 0.1$? Are the results consistent with the conclusions drawn from the ROC curves in task 1?

Table 1: Posterior probabilities for the positive class obtained by two classifiers, $M_1$ and $M_2$, on 10 test instances.

| ID | True Class | $P(Y = + \mid X, M_1)$ | $P(Y = + \mid X, M_2)$ |
|---|---|---|---|
| 1 | $-$ | 0.35 | 0.80 |
| 2 | $-$ | 0.05 | 0.40 |
| 3 | $+$ | 0.70 | 0.10 |
| 4 | $+$ | 0.45 | 0.25 |
| 5 | $+$ | 0.40 | 0.05 |
| 6 | $-$ | 0.30 | 0.15 |
| 7 | $+$ | 0.80 | 0.60 |
| 8 | $-$ | 0.15 | 0.20 |
| 9 | $+$ | 0.65 | 0.45 |
| 10 | $-$ | 0.60 | 0.30 |

5. Let the cost matrix in Table 2 be given. The total cost $C$ of a model $M$ can be calculated as:

$$C(M) = p \cdot (1 - \mathrm{TPR}_M) \cdot c_{\mathrm{FN}} + (1-p) \cdot \mathrm{FPR}_M \cdot c_{\mathrm{FP}} + p \cdot \mathrm{TPR}_M \cdot c_{\mathrm{TP}} + (1-p) \cdot (1 - \mathrm{FPR}_M) \cdot c_{\mathrm{TN}}.$$

Calculate the total cost for the models in Figure 1 for $p = 0.1$. Which model achieves the lowest cost?

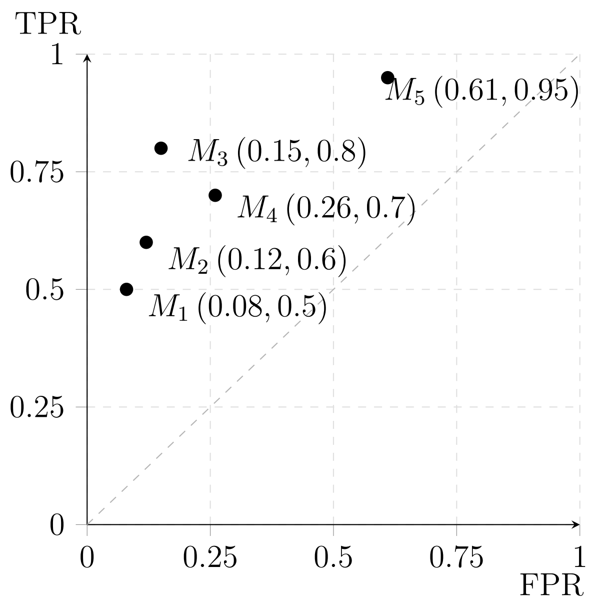


Figure 1: ROC points of five model candidates.

Table 2: Cost matrix.

| | Classifier assignment $-$ | Classifier assignment $+$ |
|---|---|---|
| True class membership $-$ | $c_{\mathrm{TN}} = 0$ | $c_{\mathrm{FP}} = 2$ |
| True class membership $+$ | $c_{\mathrm{FN}} = 10$ | $c_{\mathrm{TP}} = -1$ |